Commensurate graphite friction at complementary twist angles: a planar harmonic carrier and a test of residual dissipation

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Abstract

Graphite friction peaks at 21.8° and 38.2° have weak reported load dependence relative to the coaligned interface. Their angular pairing follows exactly from the ordinary $(2,1)$ triangular coincidence lattice. We identify a two-dimensional harmonic translation space in its seven-cell honeycomb quotient and formulate a conditional Unified Dynamics (UD) source-selection hypothesis for additional dissipation. The topology fixes neither a force magnitude nor a damping rate. Reanalysis of published first-principles line scans gives a first-harmonic, ideal static shear-strength ratio of 2.104 between separations 3.4022 and 3.34 Å, conditional on the two-dimensional potential-energy model; this is not a kinetic friction ratio. All 250 input geometries and 221 available output geometries in six audited archives sample one lateral line. Symmetry-compatible perturbations leave that line unchanged while altering transverse saddle stability. We give a source-matched off-line calculation and a response measurement that would bound the conventional baseline before a signed UD residual is tested. No parameter-free UD prediction of the observed friction or its load slope presently follows. **Keywords:** structural superlubricity; graphite; commensurate moiré; discrete Hodge theory; friction; model identifiability.

1 Observation and geometric null

Chen *et al.* report sharp friction peaks at the complementary twist angles 21.8° and 38.2°, with negligible reported load dependence relative to the coaligned state; other tested commensurate (m,n) patterns have no detectable signal [1]. A peak is a small residual force, not identically zero friction. The accessible abstract supplies no force table, uncertainty, speed-resolved trace, or measured mode linewidth for a numerical kinetic residual.

Let a_1, a_2 be equal primitive vectors separated by 60°. The two lattice directions $p = 2a_1 + a_2$ and $q = a_1 + 2a_2$ satisfy $|p|^2 = |q|^2 = 7a^2$ and $p \cdot q = 13a^2/2$. Hence

$$\begin{aligned} \theta &= \cos^{-1}(13/14) = 21.7867893\ldots^\circ, \\ 60^\circ - \theta &= 38.2132107\ldots^\circ. \end{aligned} \tag{1}$$

The 60° sum is the symmetry of the triangular Bravais lattice. Conventional moiré registry physics already accounts for the angular locations; it cannot be counted again as a UD-specific result. The coincidence index is seven and the ideal honeycomb quotient contains fourteen atoms per layer.

2 A geometric translation port

On a periodic seven-hexagon honeycomb quotient, the cellular counts are $(V,E,F) = (14,21,7)$. The oriented boundary operators obey $B_1 B_2 = 0$ and have ranks 13 and 6, so

$$\dim H^1 = 21 - 13 - 6 = 2, \quad H^1 = \ker B_1 \cap \ker B_2^T. \tag{2}$$

For the unit-edge metric, an explicit embedding M of the two lateral coordinates obeys $B_1 M = 0$, $B_2^T M = 0$, and $M^T M = \frac{7}{2} G$, where $G = \left(\begin{smallmatrix} 1 & 1/2 \\ 1/2 & 1 \end{smallmatrix}\right)$ is the planar lattice Gram matrix [5]. This is a geometric encoding of ordinary relative translation. A closed unit-metric Hodge–Dirac generator annihilates that harmonic sector; it supplies neither interlayer energy nor Ohmic drag.

A possible UD source restriction is more specific. Assign edge weights $W = \mathrm{diag}(w_e)$ and define the projected incidence-source map

$$S(w) = M^T W B_1^T. \tag{3}$$

For a uniform weight $w_e = w$, $S(w) = w(B_1 M)^T = 0$. In a seven-orbit construction, its isotropic quadratic strength can be written $\alpha(w) = \frac{1}{4}\sum_{i<j} c_{ij}(w_i - w_j)^2$ with $c_{ij} \geq 0$ [5]. Thus source contrast is necessary for *this* selected port. The physical edge weights and their coupling to phonons are additional, measurable constitutive inputs. Ordinary heterogeneous interfaces can obey the same symmetry, so the zero rule alone is not distinctive evidence for UD.

3 Material response and conventional baseline

Let x denote two lateral translations and h internal coordinates. A stable local energy has lateral block H_{xx}, internal block $K \succ 0$ and mixed block C. Relaxation and an independently established passive internal Rayleigh matrix $D_h \succeq 0$ give

$$H_{xx}^{\mathrm{rel}} = H_{xx} - C K^{-1} C^T, \tag{4}$$

$$\Gamma_{\mathrm{add}} = C K^{-1} D_h K^{-1} C^T \succeq 0. \tag{5}$$

These are conditional linear-response relations. A finite closed model has $D_h = 0$ and no DC Ohmic coefficient. At a threefold-fixed registry, a scalar height cannot bilinearly couple to the lateral two-dimensional representation; a height doublet can. Nothing in Eq. (2) fixes C, K, D_h, the sliding contact, or a force at finite speed.

A published Kolmogorov–Crespi (KC) calculation and a vdW-DF3 first-principles study disagree strongly on the $(2,1)$ shear scale: a registered KC-plus-plate example gives $8.73\,\mathrm{cm}^{-1}$ for two mobile layers, while vdW-DF3 reports 0.404–0.590 cm^{-1} across distinct relaxation protocols [2, 3, 5]. Their separations and mechanical constraints also differ. The example is therefore a sensitivity demonstration, not a graphite calibration of Eq. (5).

Within the type-II *first-harmonic* potential used in the vdW-DF3 analysis, the ideal rigid shear frequency $\widetilde{\nu}$ and static minimum-path shear strength τ obey the amplitude-free relation

$$\frac{\tau}{\widetilde{\nu}^2} = 8.48565 \times 10^4 \ \frac{\mathrm{Pa}}{(\mathrm{cm}^{-1})^2} \quad (N_c = 7,\ a = 2.4660\,\text{Å}). \tag{6}$$

For fixed a and unchanged registry branch, $d\ln\tau/dP = 2\,d\ln\widetilde{\nu}/dP$; if a changes, subtract $d\ln a/dP$. This tests one conventional static model, not kinetic friction. The relation is undefined at a branch inversion where the harmonic amplitude crosses zero [3, 5].

4 What the first-principles archive determines

The public vdW-DF3 lateral scans at 3.4022 and 3.34 Å have 37 completed points each. Replaying their full profiles yields $U_1 = 0.0793397$ and $0.166933\,\mu$eV per upper atom. Their first-harmonic ideal static strengths are 1.3866×10^4 and 2.9175×10^4 Pa, a ratio

2.1040 for an ideal rigid compression to roughly 0.72–0.74 GPa [3, 4, 5]. It is a finite model ratio, not a local derivative or a measured load response of the friction peaks.

The six SHA-checked archived sets contain 250 input geometries and 221 available XML output geometries, but all first-carbon lateral markers are collinear with one high-symmetry line to floating precision [5, 4]. No sampled energy directly measures its transverse curvature. In first-shell phases $u = k_x x'$, $v = k_y y'$, a sixfold-compatible perturbation

$$F(u,v) = [1-\cos(u-3v)][1-\cos(u+3v)][1-\cos(2u)] \quad (7)$$

vanishes on the entire sampled line $u = 3t$, $v = 2\pi/3 + t$ and leaves the AB local Hessian unchanged. At the line's candidate saddle, the transverse Hessian of $U_1 f + \varepsilon F$, with $f = 2\cos v \cos u + \cos 2v$, is $k_y^2(6U_1 + 48\varepsilon)$. Its sign changes at $\varepsilon/U_1 = -1/8$. This is an algebraic non-uniqueness example, *not* an inferred graphite harmonic. Even a finite set of off-line energy and force probes cannot uniquely fix the infinitesimal Hessian without a declared Fourier or smoothness bound [5].

5 A discriminating measurement sequence

Static PES control. Recalculate the published 3.4022 Å index-30 configuration and four source-matched positions with the fourteen upper atoms shifted by ± 0.04 and ± 0.08 Å along $n = (1/2, -\sqrt{3}/2)$. Use matched pseudopotential, cutoffs, k mesh, vacuum, and full-precision QE XML energy. With $D(h) = E(h) + E(-h) - 2E(0)$ per upper atom, the fitted first-shell model predicts $D(0.08\,\text{Å}) = +0.0450423\,\mu$eV. The illustrative $\varepsilon = -0.2U_1$ in Eq. (7) predicts $-0.0127457\,\mu$eV. A negative resolved contrast establishes an off-line energy decrease at that finite step. The actual calculation has not been run. The two-step local-Hessian extrapolation requires a validated higher-derivative bound and correlated-error accounting; at a hypothetical independent per-energy uncertainty of $0.001\,\mu$eV, the example's extrapolated negative curvature is only 2.17 standard deviations from zero before truncation error [5].

Dynamic loss control. At each angle and pressure, measure a background-subtracted small-amplitude complex lateral force response about a single registry, plus the mode-resolved internal stiffness and linewidths. The model's quadrature force and loss per cycle at drive amplitude A and angular frequency ω are $|F_Q| = 14A\omega\Gamma_{\text{add}}$ and $W_{\text{cycle}} = 14\pi A^2\omega\Gamma_{\text{add}}$ for one ideal 14-upper-atom cell. These are local oscillatory observables, not a steady sliding-force prediction. Contact edges, substrate loss and direct lateral damping must be included in the ordinary baseline.

Held-out UD test. Before viewing paired-angle kinetic-force curves, source the edge weights and the material bath independently. A restrictive candidate is $C = S(w)R$, with independently specified mode map R. Equations (3) and (5) then imply

$$\Gamma_{\text{port}} = S(w)\mathscr{L}S(w)^T, \quad \mathscr{L} = RK^{-1}D_hK^{-1}R^T \succeq 0. \quad (8)$$

It predicts a zero contribution for uniform source weights and a positive-semidefinite tensor for an additive passive channel. Freeze S, R, K, D_h, contact controls and a finite-speed force protocol using independent data. Only then predict the *signed* residual $F_{\text{obs}} - F_{\text{conventional}}$ and its load derivative at both complementary angles. A flexible conventional bath can realize the same positive tensor; Eq. (8) becomes specifically UD-testable only through source-fixed, out-of-sample restrictions that the conventional fit does not inherit. Failure of the source span, loss spectrum, sign, or held-out paired-angle forecast rejects or narrows this material bridge.

6 Conclusion

The 60° pairing selects a planar triangular/honeycomb description but is already explained by lattice geometry. The finite carrier supplies a precise translation port and a conditional source-selection rule. The first-principles line scans constrain a model-dependent static PES, while transverse stability, material loss and the measured finite-speed load response remain independent questions. A UD explanation of the remaining friction is a falsifiable hypothesis only after the ordinary two-dimensional PES, contact dynamics and loss spectrum are bounded and the proposed source rule makes a signed held-out prediction. Present evidence does not supply that prediction.

Data and reproducibility

The numerical replays and exact finite constructions are retained as versioned scripts, inputs and checksum manifests in the UD repository draft series [5]; the original QE data are archived separately [4]. No experimental force table or new off-line DFT result is included. Constructor verification is not independent review.

References

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[3] I. V. Lebedeva *et al.*, "Robust structural superlubricity of twisted graphene bilayer and domain walls between commensurate moiré pattern domains from first-principles calculations," arXiv:2511.01338 (2025), Tables 2, 4, 5, and 6.

[4] I. V. Lebedeva *et al.*, first-principles source data, Zenodo DOI 10.5281/zenodo.16318186.

[5] UD research record, graphite captures and draft PRs #445, #446, and #447. Exact constructions, source checksums, scripts, and working paper v9; development analyses, not independent validation.